

SYSTEM VULNERABILITY ASSESSMENT USING NMAP

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Tool Used: Nmap (Network Mapper)

Platform: Kali Linux Virtual Machine (VirtualBox)

INTRODUCTION

A vulnerability assessment is the process of identifying potential security weaknesses in a computer system or network. Cybersecurity professionals use specialized tools to scan systems and detect open ports, running services, and possible vulnerabilities. Nmap (Network Mapper) is one of the most widely used tools for network discovery and security auditing. In this task, Nmap was used to analyze the local system to identify open ports and services that could represent potential security risks.

OBJECTIVE

The objective of this project is to perform a vulnerability assessment on the local machine using Nmap and identify any open ports or running services that may pose potential security risks.

TOOLS AND ENVIRONMENT

The following tools and environment were used to perform the vulnerability assessment:

- Kali Linux (running on VirtualBox)
- Nmap (Network scanning tool)
- Terminal interface

METHODOLOGY

Step 1 – Verify Nmap Installation

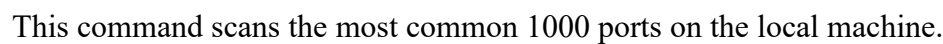
First, the Nmap tool was verified using the command below to ensure that the scanning tool was properly installed.

`nmap --version`



A basic scan was performed on the localhost to detect commonly used open ports.

```
nmap 127.0.0.1
```



The scan showed that the host was active, but all scanned ports were closed.

A service version detection scan was performed using the following command.

```
nmap -sV 127.0.0.1
```

Result:

Step 4 – Operating System Detection

```
nmap -O 127.0.0.1
```

Result:

Step 5 – Aggressive Scan

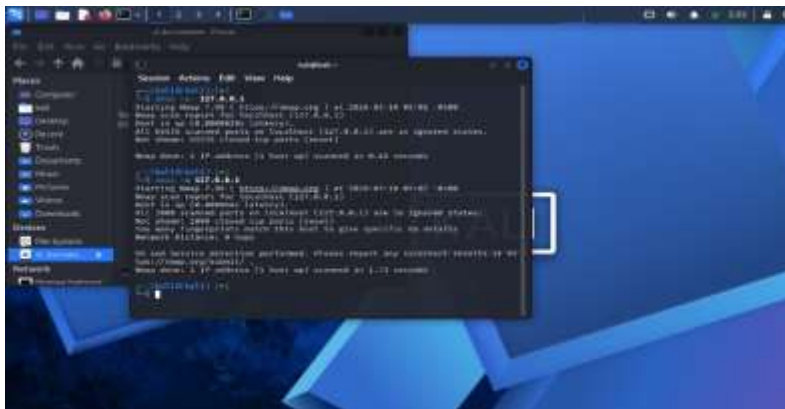
```
nmap -A 127.0.0.1
```

- OS detection
- Version detection

- Script scanning
- Traceroute

Result:

The scan confirmed that the host was active, but no open ports were detected.



RESULTS

The vulnerability assessment scan revealed the following findings:

- The host system was active.
- No open ports were detected.
- All scanned ports were closed.
- No active services were running on the scanned ports.

This indicates that the system has minimal network exposure and a reduced attack surface.

SECURITY ANALYSIS

Closed ports indicate that the system is not exposing network services that could be exploited by attackers. Systems with fewer open ports generally have a lower risk of remote network attacks. However, regular vulnerability scanning and monitoring are recommended to ensure that no unauthorized services become active.

CONCLUSION

The vulnerability assessment using Nmap successfully analyzed the local machine for open ports and running services. The results showed that all scanned ports were closed, indicating a secure configuration with minimal exposure to network-based threats. This task demonstrated how Nmap can be used effectively for basic vulnerability scanning and security analysis.